The Information Age

The development of the first programmable computers in the mid-20th century launched what is known as the Information Age. The first computers were huge and expensive but by the 1980s scientists had developed small, low-cost computers that people could use in their homes. By the 1990s, the invention of the World Wide Web had begun to transform the way we communicate, study, and work.

Electronic devices can turn information into electrical signals and back again. Two inventions in the mid-20th century powered electronics forward. All electronic devices are based on the transistor—a switch that controls the flow of an electrical signal, turning it on or off. The creation of the microchip, which can store millions of transistors on one tiny piece of material, made it possible to produce electronic gadgets that were much smaller, cheaper, and more reliable. By the 1980s, the electronic revolution was well underway as computers became cheaper to produce and more people were able to buy them for use at home, work, and in schools.

Today, technology makes it possible for us to communicate across long distances and gives us access to a world of information on the internet at the touch of a button. Yet some worry that we have become too reliant on our devices, and that cutting down on face-to-face communication has led to us feeling more isolated.

ALAN TURING

Alan Turing (1912–1954) was a British mathematician and a computing pioneer. In 1936, he came up with the idea of a machine that could solve any mathematical problem given the right instructions. He never built such a "Turing machine," but the idea led to the modern computer.

HISTORY OF COMPUTERS

1822 British inventor Charles Babbage designs a machine called the Difference Engine, which is able to perform difficult calculations.

1843 British mathematician Ada Lovelace works on Charles Babbage's new Analytical Engine, an ancestor of modern computers. She is the first computer programmer.

ADA LOVELACE

1946 US engineers develop the ENIAC (Electronic Numerical Integrator and Computer), the first programmable computer. It can perform 5,000 calculations per second.

1975 The Cray-1 is invented. This "supercomputer" can perform 240 million calculations per second. It is used to design nuclear weapons and for weather forecasting.

1982 The Commodore 64 is released, and becomes the biggest-selling home computer of all time. It opens up the world of video games, word processing, and spreadsheets to millions of people.

◀ **PIONEERING WOMEN**
Many of the early pioneers in the field of computer programming were women. Black American computer programmer Melba Roy Mouton led a group of NASA scientists, known as "computers," in their astronomical calculations. Another US computer scientist Grace Hopper (left) made a key contribution to the development of computer languages.